JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
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An Autonomous Institute, with NAAC "A" Grade
Department of CSE
"A Place to Learn ; A Chance to Grow"
Session 2024-25

CLASS TEST 2 EXAMINATION

Course: B. Tech. in Computer Science & Engineering
Subject: Operating System
Subject Code: CS3T002
Year/Semester: 3rd SEM CSE
Max Marks: 20 **Date:04/ 10/2024** **Duration:- 1 Hr.**

Instructions to the Students:
1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students will be able to:
1 Remember the concepts, core structure of Operating Systems.
2 Understand the process management policies and scheduling of processes by CPU.
3 Evaluate the requirement for process synchronization and coordination handled by operating system.
4 Describe and analyze the memory management and its allocation policies.
5 Analyze various device and resource management techniques for timesharing

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Swapping.	L1/CO4	1
Q.2	State logical address Space.	L1/CO4	1
Q.3	Enlist Page replacement policy.	L1/CO5	1
Q.4	Define Page fault.	L1/CO5	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q.5	Distinguish between Logical Address Space and Physical Address Space.	L4/CO4	2
Q.6	Illustrate the disadvantage of fixed partitioning and explain how it will be solved.	L2/CO4	2
Q.7	State Resource Allocation Graph with example	L1/CO3	2
Q.8	Compare internal and external fragmentation.	L2/CO4	2

SECTION C

1. **Solve any two questions.**

Q.9	Explain Paging with Example	L1/CO5	4

Q.10	Apply the bankers algorithm by considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t0 following snapshot of the system has been taken: Q.1: What will be the content of the Need matrix? Q.2: Is the system in a safe state? If yes, then what is the safe sequence?	L3/CO3	4

Process	Allocation	Max	Available
	A B C	A B C	A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

Q.11	Consider the page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 3 with 4 page frames. Find number of page faults Using Least Recently Use page replacement strategy.	L3/CO5	4

********** **********All the best********** **********